UChicago Stat 32960

Topics in dimension reduction

Phillip Nicol

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1 Introduction

2 Principal Component Analysis

2.1 Population and sample principal components

Definition 2.1. Let $X \in \mathbb{R}^p$ be a random vector with mean $\mu = \mathbb{E}[X]$ and covariance matrix $\Sigma = \mathrm{Cov}(X)$. The **first population principal component direction** is a unit vector v_1 such that

$$v_1 \in \arg \max_{v \in \mathbb{R}^p, \|v\|_2 = 1} \mathrm{Var}(X^\top v) \tag{1}$$

Calculation that this is $v^\top \Sigma v$ maximization.

The **second population principal component direction** is a unit vector v_2 that maximizes the variance of $X^\top v$ subject to the constraint that v_2 is orthogonal to v_1:

$$v_2 \in \arg \max_{v \in \mathbb{R}^p, \|v\|_2 = 1, v^\top v_1 = 0} \mathrm{Var}(X^\top v) \tag{2}$$

In general, blah

The k-th population principal component score is defined as the random variable

$$Z_k = (X - \mu)^\top v_k \tag{3}$$

Proposition 2.2. *Suppose $X \in \mathbb{R}^p$ is a random vector with mean μ and covariance matrix Σ. Suppose that the eigenvalues of Σ are distinct and ordered as $\lambda_1 > \lambda_2 > \cdots > \lambda_p > 0$ with corresponding eigenvectors $v_1, v_2, \ldots, v_p$. Then the population principal component directions are given by the eigenvectors of Σ, and the population principal component scores satisfy $\mathrm{Var}(Z_k) = \lambda_k$ for $k = 1, 2, \ldots, p$ and $\mathrm{Cov}(Z_k, Z_j) = 0$ for $k \neq j$.*

Proof. TODO □

EXAMPLE with $\mathcal{N}(0, \Sigma)$

Statistical interpretation (signal plus noise)

Sample PCs

3 Singular Value Decomposition

4 Random Matrix Theory